Econometrics I

Lecture 5b: Causal Diagrams (DAGs)

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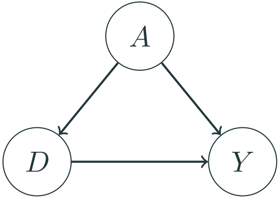
Fall 2026

Two languages for the same problem

- Potential outcomes (last hour) tell us *what we want*: $\mathbb{E}[Y_i(1) - Y_i(0)]$, and what randomization buys us.
- Causal diagrams (DAGs) encode *what we believe*: which variables cause which, drawn as arrows. The assumptions are the *missing* arrows.
- Why learn both?
  - DAGs make "what should I control for?" a mechanical question with a right answer — and expose when a control makes things worse.
  - Potential outcomes handle heterogeneity and identification-by-design more gracefully. Economists mostly publish in this language.
- Readings: Mixtape Ch. 3; Hünermund & Bareinboim (2021); Cinelli, Forney & Pearl (2022), "A Crash Course in Good and Bad Controls."

What is a DAG?



D = education, Y = wages, A = ability

- Nodes are variables; arrows are direct causal effects; no cycles allowed (“acyclic”).
- Every missing arrow is an assumption: this graph asserts nothing else causes both D and Y.
- Unobserved variables (like ability) still belong in the graph — being unobservable doesn’t stop something from causing things.

Paths: how association travels

In the graph on the last slide there are two paths from D to Y:

1. $D \rightarrow Y$: the causal path. This is the thing we want to measure.
2. $D \leftarrow A \rightarrow Y$: a backdoor path. Association flows along it even though D causes none of it.

- The regression of Y on D alone picks up *both* paths — this is exactly the omitted variables bias formula from Lecture 2, drawn as a picture.
- Identification strategy = a plan for shutting every backdoor path while leaving the causal path alone.

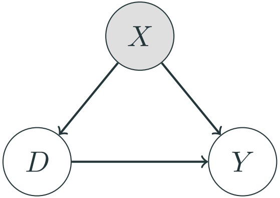
The grammar: three ways association flows

Every path, however long, is built from three pieces (Mixtape Ch. 3):

Motif	Shape	Default	Conditioning on the middle node
Chain	$A \to B \to C$	open	blocks it
Fork	$A \leftarrow B \to C$	open	blocks it
Collider	$A \to B \leftarrow C$	blocked	opens it

- Association flows between two variables if any path between them is open. A path is open unless it is blocked at some node along the way.
- Everything on the coming slides — good controls, bad controls, selection bias — is just these three rows applied to a picture. (The formal name for the full rule is *d-separation*.)
- The collider row is the one that breaks intuition: it is the only place where controlling for something *creates* association.

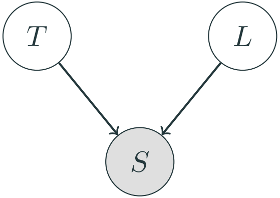
Good controls: closing a backdoor



Shading $=$ conditioned on.

- Conditioning on a confounder X (regression control, matching, stratification) blocks the backdoor path $D \leftarrow X \rightarrow Y$.
- This is the backdoor criterion (informally): find a set of observed variables that blocks every backdoor path, condition on them, and the remaining association is causal.
- “Selection on observables” (matching, next month) is exactly the claim that such a set exists *and you have it.*

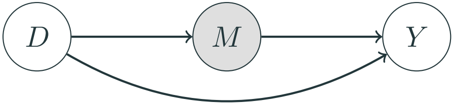
Colliders: conditioning can *open* a path



T = talent, L = luck, S = startup gets funded.

- S is a collider on the path $T \to S \leftarrow L$: two arrows collide into it. A collider blocks the path *by default*.
- But conditioning on a collider opens the path: among *funded* startups, talent and luck are negatively correlated — if a funded founder wasn't talented, they must have been lucky.
- Talent and luck are independent in the population. The correlation is manufactured by the conditioning. This is selection bias, drawn as a picture.

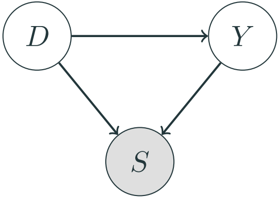
Bad control #1: the post-treatment variable



D = education, M = occupation, Y = wages

- M is a mediator: part of *how* education raises wages is by changing occupations.
- Controlling for occupation blocks part of the effect you are trying to measure. The coefficient on D is no longer the total effect.
- Rule of thumb with real teeth: do not control for variables that are themselves outcomes of the treatment. When in doubt, ask: was this variable determined before or after D?

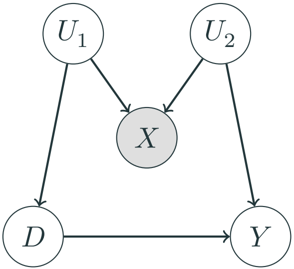
Bad control #2: the collider / sample-selection control



D = ad exposure, Y = purchase intent, S = visits the site

- Estimating the ad effect only among site visitors conditions on a collider: both seeing the ad and wanting the product cause visits.
- Within visitors, exposure and intent are spuriously related even if the ad does nothing.
- This is the same disease as Heckman's sample selection (Lecture 11) and the "conditioning on graduation" problem in education studies. Who is in your sample, and did the treatment help put them there?

M-bias: even a pre-treatment control can hurt



U_1, U_2 unobserved; X observed before treatment.

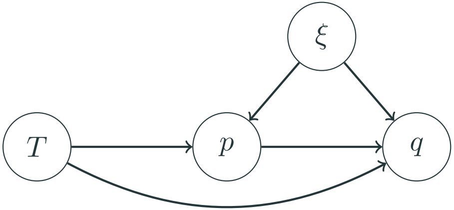
- The comforting heuristic "pre-treatment controls are always safe" is false. Here X is measured before D — and it is a collider on the path $D \leftarrow U_1 \rightarrow X \leftarrow U_2 \rightarrow Y$.
- Unconditionally that path is blocked at X; controlling for X opens it, importing bias where there was none.
- Story: D = adopting analytics software, Y = productivity, X = a pre-adoption "tech enthusiasm" survey score driven by managerial ambition (U_1, which also drives adoption) and workforce skill (U_2, which also drives productivity).
- Timing is not enough — you have to think about *why* the control varies. (Cinelli, Forney & Pearl

A field guide to controls

Candidate control X	Control for it?	Why
Common cause of D and Y (confounder)	Yes	closes the backdoor
Proxy for an unobserved confounder	Usually yes	closes it partially
Pre-treatment predictor of Y only	Yes (precision)	shrinks SEs, no path to D
Mediator ($D \to X \to Y$)	No*	blocks the effect itself
Collider (caused by two paths' worth of causes)	Never	opens a closed path
Descendant of Y	Never	conditioning on the outcome
Instrument (affects D only)	No	amplifies remaining bias

- Row 3 is the positive surprise: in an A/B test, adjusting for *pre-experiment* outcomes (last month's spend) is pure precision gain — this is what platform experimentation teams do routinely.
- *Unless the direct effect is your estimand and you have an instrument for the mediator — the case study coming next.
- The full 18-model catalog: Cinelli, Forney & Pearl (2022), "A Crash Course in Good and Bad Controls."

Case study: when prices respond to your experiment



T = randomized ad/feature, p = price, q = quantity, ξ = demand shock (unobserved)

- A marketplace A/B test improves product pages. Demand rises — and treated sellers raise prices, offsetting part of the sales gain.
- Often the estimand is the direct effect: the demand-curve shift, *holding price constant*.
- So this time we *want* to hold the mediator fixed — but p is a doubly bad control: post-treatment ($T \to p$) *and* correlated with the demand shock ($\xi \to p$: sellers reprice when demand is hot).

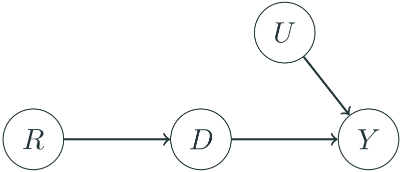
Three regressions, one instrument

With randomized T, positive direct effect, and prices that respond (Jeziorski, Leng & Seiler 2025, working paper):

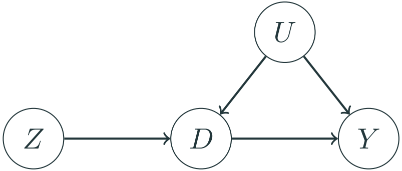
1. q on T: the total effect — demand shift *minus* the price offset. A fine estimand, but it is not the demand shift.
2. q on T and p: biased for the direct effect even though T is randomized — p drags ξ into the regression. Typically the bias shrinks but does not vanish, and it has the same sign as omitting p. The middle answer is still a wrong answer.
3. The fix: a price instrument (a cost shifter that moves p but not demand) recovers the direct effect. Instruments are next week's topic — note that here the instrument is for the control, not the treatment.

- Built-in diagnostic: regress p on T. If price responds to your randomized treatment, regressions (1) and (2) answer different questions than you think.
- Stakes: in supermarket scanner data, not instrumenting price overstates feature-advertising

The research designs of this course, as DAGs



Randomization: coin flip R causes D; no arrow into D from U.



Instrumental variables: the missing arrows $Z \to Y$ (exclusion) and $U \to Z$ (exogeneity) ARE the IV assumptions.

Every design we meet from here on is a claim about missing arrows. When you referee a paper, draw its DAG — the contestable assumption is usually an arrow the authors need to be absent.

You don't have to do this in your head

- dagitty (`https://dagitty.net`, also an R package): draw your DAG in the browser and it computes, automatically,
  - every minimal adjustment set that identifies your effect ("control for exactly these"), and
  - the testable implications: conditional independencies your DAG claims, which you can check in data. A DAG is a falsifiable object, not a vibe.
- Workflow worth adopting: draw the DAG *before* touching the data; put it in your appendix; let referees argue with the arrows instead of guessing your assumptions.
- Pearl's machinery goes much further — front-door identification, do-calculus, combining experimental and observational data ("data fusion") — see Hünermund & Bareinboim (2021) if you want the full toolkit. For this course, backdoors and colliders do 95% of the work.

Where DAGs strain (and economists push back)

- Simultaneity: supply and demand determine price and quantity *jointly* — a cycle, which a DAG cannot draw. Our oldest identification problem (Working 1927, Lecture 6) sits awkwardly in this language.
- Heterogeneous effects: LATE, compliers, and monotonicity (Lecture 7) are natural in potential outcomes and clumsy in DAGs.
- Where DAGs shine: bad controls, sample selection, and transparent communication of assumptions — especially in fields (marketing, IS) where "we controlled for everything" is still treated as a virtue.
- Verdict: use the DAG to decide *what to control for*; use potential outcomes to decide *what your estimate means*.

When units interfere

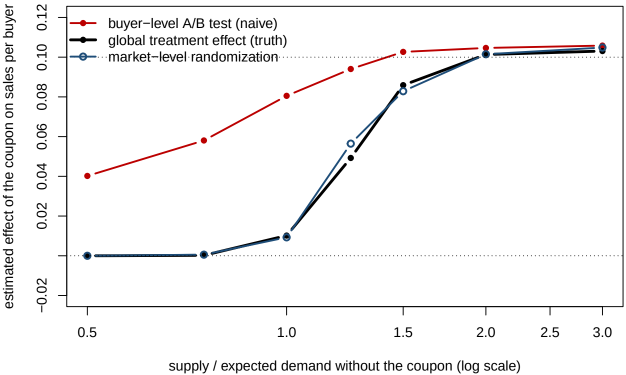
The assumption every A/B test makes without saying so

- Everything so far assumed my treatment does not affect *your* outcome. That is the no-interference half of SUTVA. As a DAG: no arrows between units.
- It fails whenever units share something:
  - a market: a coupon for treated buyers lets them grab inventory control buyers wanted; promoting treated sellers diverts demand from control sellers;
  - a network: a feature shown to treated users changes what their control friends see;
  - a resource: discounted rides for treated riders lengthen waits for control riders; drivers reallocate toward treated neighborhoods;
  - geography or time: a store-level promotion pulls shoppers from the control store across the street.
- When it fails, "treated minus control" is not the effect of treating everyone. It is the effect of treating *half* the units *while the other half is untreated*, which is an experiment nobody plans to run at scale.

A marketplace, simulated

- 200 markets, 200 buyers each, a shared inventory per market. A buyer wants to purchase with probability 0.20; a coupon raises that to 0.30. Purchases happen in random order until inventory runs out.
- Three numbers, each with a known truth:
  - Buyer-level A/B: randomize the coupon across buyers within each market; compare treated and control buyers. The experiment every platform runs by default.
  - Global treatment effect: sales per buyer with the coupon for everyone minus sales per buyer with it for no one. The number the business decision needs.
  - Market-level randomization: give the coupon to everyone in a random half of the markets.
- The one knob: how much inventory there is relative to demand without the coupon.

When supply binds, the A/B test measures the wrong thing



With inventory at 75% of demand, the coupon cannot raise total sales at all: the truth is zero. The buyer-level test reports $+0.06$, because treated buyers took units that control buyers would have bought. Only when supply is slack do the three numbers agree. Blake and Coey (2014) found exactly this at eBay.

The same picture as a table

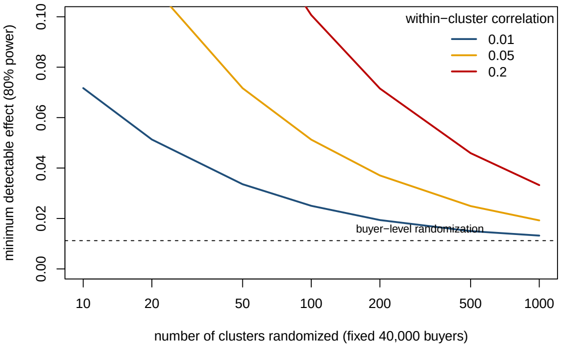
Supply / control demand	Buyer-level A/B	Global effect (truth)	Market-level randomization	SE, buyer-level
0.75	0.058	0.000	0.001	0.0034
1.00	0.081	0.010	0.009	0.0038
2.00	0.105	0.101	0.101	0.0045

- The buyer-level test is precise and wrong: a tight standard error around a number that answers no business question. A partially binding constraint biases it partway.
- Market-level randomization is unbiased for the global effect; its price is precision, since the effective sample size is the number of markets. Johari, Li, Liskovich and Weintraub (2022) derive the bias in a queueing model and show that randomizing over buyers *and* listings reduces it.

Designs that survive interference

- Cluster randomization. Randomize at the unit where spillovers stop: markets, cities, stores, ego-networks (Saveski et al. 2017 at LinkedIn). Estimand: the global effect, if clusters are closed.
- Switchbacks. One market, treatment on and off in alternating time blocks (DoorDash, Lyft; Bojinov, Simchi-Levi and Zhao 2023). For markets you cannot split; needs carryover to die out between blocks.
- Two-sided designs. Randomize buyers and listings jointly so treated-buyer, treated-listing cells exist (Johari et al. 2022; Bajari et al. 2023 at Amazon).
- Budget-split. Separate inventories or budgets for treatment and control so they cannot compete (ad auctions: Liu et al. 2021 at LinkedIn). Removes the interference instead of averaging over it.
- Exposure mapping. Each unit's treatment is a function of what its neighbors got; estimate effects by exposure level (Aronow and Samii 2017). The general framework the others are cases of.

What cluster randomization costs: back to power



The design effect $1 + (m - 1)\rho$, with m buyers per cluster and within-cluster correlation ρ. Fixing $40{,}000$ buyers, the minimum detectable effect rises as clusters get fewer and bigger: 50 metros at $\rho = 0.05$ detect roughly four times less than buyer-level randomization would. Plan power at the cluster level.

How to tell whether interference is biting

- Ask where the outcome comes from. If treated and control units draw on the same inventory, budget, drivers, or attention, assume interference until shown otherwise.
- Run both designs on a small scale. A buyer-level and a market-level test of the same change should agree if there is no interference (Saveski et al. 2017 formalize this as a test).
- Vary the treated share. In a buyer-level test, randomize the *fraction* treated across markets (10%, 50%, 90%). If control buyers' outcomes move with the treated share, they are being affected (Hudgens and Halloran 2008).
- Look at the control group's trend. If control outcomes fall when treatment starts, the "effect" is partly displacement.
- Report which estimand you have. "Treated minus control in a 50% buyer-level test" is a fine sentence. "The effect of the coupon" is not, unless the design justifies it.

- A DAG is a set of causal assumptions; the missing arrows do the work.
- Three motifs — chain, fork, collider — are the whole grammar. Backdoor paths are confounding; blocking them is what "controlling for" means.
- Colliders reverse the logic: conditioning *creates* association. Selection into the sample is a collider, and M-bias shows even pre-treatment controls can be colliders.
- Bad controls: ask *why* the variable varies, not just *when* it was measured. When in doubt, the field guide — or let dagitty compute the adjustment set.
- Every design in this course is a claim about missing arrows — draw the picture before you believe the number.

Let’s look at the workbook now.

- Open `inclass_session5.Rmd` (in `Fall/L7 - Program Evaluation and Selection` in the course repo).
- Reproduce the power calculations, the ten-line randomization inference, the funded-startup collider, and the marketplace interference simulation.
- Then work the three problems with a neighbor:
    1. The bad control, live
    2. Fix the marketplace experiment
    3. Design your own experiment

Thanks!